

Sequential administration of efgartigimod shortened the course of Guillain–Barré syndrome: a case series

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Abstract: Guillain–Barré syndrome (GBS) is a serious neurological condition with limited treatment options. A recent report demonstrated successful treatment with efgartigimod alone in two patients with GBS, although it did not significantly shorten the disease duration. This case series investigates the effects of sequential efgartigimod administration in patients with different GBS phenotypes and varying levels of disease severity. All three patients tested positive for immunoglobulin G (IgG) antibodies against serum gangliosides. In Case 1, the patient was treated with 0.4 g/kg of intravenous immunoglobulin (IVIg) for 5 days, showing minimal recovery. After receiving 3 weekly doses of efgartigimod (10 mg/kg), the patient achieved independent ambulation 19 days post-onset, with a reduction in serum ganglioside antibody titers and total IgG levels. Case 2 involved a middle-aged man with Miller Fisher syndrome (MFS)-GBS overlap, who experienced worsened autonomic dysfunction following IVIg treatment. After three doses of efgartigimod, the patient showed symptom improvement within 1 month, alongside a reduction in IgG antibody levels. In Case 3, a 27-year-old male with MFS-GBS overlap, initially unresponsive to IVIg, showed significant improvement in ophthalmoplegia following two doses of efgartigimod, with his serum ganglioside antibodies eventually becoming undetectable. Our findings suggest that sequential efgartigimod treatment may effectively reduce serum anti-ganglioside antibody titers and potentially shorten the disease course in severe GBS and MFS-GBS overlap syndrome. Additionally, it may offer clinical benefits for patients with GBS who have a poor or no response to IVIg, particularly in treating ophthalmoplegia.

Keywords: Disease course, efficacy, efgartigimod, Guillain–Barré syndrome, Miller Fisher syndrome

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Introduction

Guillain–Barré syndrome (GBS) is a leading cause of acute flaccid paralysis and is a severe, IgG-mediated peripheral neuropathy. It is typically classified into two subtypes: acute inflammatory demyelinating polyneuropathy, and acute motor axonal neuropathy (AMAN).^{1–6} Approximately 20%–30% of patients with GBS in the acute phase develop acute respiratory failure, requiring mechanical ventilation.⁴ Miller Fisher syndrome (MFS), a variant of GBS, presents with a classic triad of ophthalmoplegia, ataxia, and areflexia.⁷ The median recovery times

for ataxia and ophthalmoplegia have been reported to be 1 and 3 months, respectively.⁸ However, in some cases, MFS may overlap with GBS (MFS-GBS overlap syndrome), which can prolong the disease course.^{9,10}

Currently, intravenous immunoglobulin (IVIg) and plasma exchange (PE) are the standard treatments for GBS.^{1,5} Despite their established efficacy, at least 20% of patients experience symptom exacerbation during or shortly after IVIg or PE, and additional rounds of IVIg are often ineffective, raising safety concerns.^{11–14} Furthermore,

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long-term symptoms, such as autonomic dysfunction, may persist in many patients even after receiving first-line treatments.^{15,16} This highlights the need for alternative therapies that are both safer and more effective in treating GBS.¹³

Efgartigimod, a novel IgG1 Fc fragment that targets the neonatal Fc receptor (FcRn), has been shown to reduce total IgG levels by 50% after a single dose, with a further reduction of nearly 75% when administered over a week. These results suggest its potential as a treatment for autoimmune conditions, such as myasthenia gravis.^{17,18} Recently, efgartigimod monotherapy was evaluated in two patients with GBS, with results suggesting possible therapeutic benefit, although the treatment did not significantly shorten the disease duration.¹⁹

The objective of this case series was to assess the clinical benefits of sequential efgartigimod treatment following first-line IVIg therapy in patients with GBS, with particular emphasis on its effects on disease severity and duration. Here, we present the treatment outcomes of three patients with distinct GBS phenotypes and varying degrees of disease severity.

Materials and methods

Patient data were extracted from the medical records of three patients with GBS treated at the

Department of Neurology, West China Hospital, Sichuan University. Motor function and activities of daily living were assessed using the Medical Research Council (MRC) scores, the GBS disability score, and the Barthel index.

Results

Table 1 provided a summary of the clinical characteristics of the three patients with GBS, while Table 2 outlined the changes in serum total IgG and ganglioside antibody levels at various time points. Prior to treatment, cerebrospinal fluid (CSF) IgG levels were elevated in all patients, exceeding the normal range (Table S4).

Case studies

Case 1

A 56-year-old male patient presented to the emergency department with progressive weakness and dyspnea, which began 14 days after a bout of diarrhea. Upon admission, the patient was in respiratory distress, with a respiratory rate of 25–30 breaths/min and oxygen saturation around 80%. Neurological examination revealed dysarthria, limited eye movement, and mild eyelid and facial muscle weakness. Distal limb muscle strength was graded 0, and proximal limb muscle strength was graded 1 on the MRC scale. Limb muscle tone was reduced, and tendon reflexes

Table 1. Clinical features of GBS patients.

Patients	Age	Sex	Neuropathy symptoms	Prodromal history	Diagnosis	Timing of IVIg administration	Timing and numbers of efgartigimod administration	Total duration (days)
Case 1	56	Male	Weakness, dyspnea	Diarrhea	Severe GBS (AMAN subtype)	Day 1 (the 3rd of onset)	Day 11 (the 13th of onset), 3	27 days
Case 2	54	Male	Eye muscle paralysis, ataxia, autonomic symptoms, peripheral sensory function impairment, weakness	Cold	MFS-GBS overlap syndrome	Day 2 of onset	Day 7 of onset, 4	27 days
Case 3	27	Male	Eye muscle paralysis, ataxia, weakness	None	MFS-GBS overlap syndrome	Day 1 (the 3rd of onset)	Day 23 (the 25th of onset), 2	32 days

‘Day 1’ refers to the first day of the patient’s hospitalization; ‘the 3rd of onset’ refers to the third day after the onset of disease. AMAN, acute motor axonal neuropathy; GBS, Guillain–Barré syndrome; IVIg, intravenous immunoglobulin; MFS, Miller Fisher syndrome.

Table 2. Changes in serum total IgG and gangliosides antibodies at different timepoints.

Patients	Onset of disease (day) and serum indicators			
	Indicators	Day 4	Day 18	
Case 1	Gangliosides antibodies	anti-GT1a IgG (++)	anti-GT1a IgG (-)	
		anti-GD3 IgG (+++)	anti-GD3 IgG (+)	
	Serum total IgG	20.2 g/L↑	6.71 g/L↓	
Case 2	Indicators	Day 2	Day 16	Day 130
	Gangliosides antibodies	Anti-GT1a IgG (++++)	Anti-GT1a IgG (++++)	Anti-GT1a IgG (++)
		Anti-GQ1b IgG (++++)	Anti-GQ1b IgG (++++)	Anti-GQ1b IgG (++)
		Anti-GD3 IgG (+)	Anti-GD3 IgG (-)	Anti-GD3 IgG (-)
		Anti-GT1b IgG (+)	Anti-GT1b IgG (-)	Anti-GT1b IgG (-)
	Serum total IgG	16.2 g/L	9.62 g/L	N/A
Case 3	Indicators	Day 1	Day 50	
	Gangliosides antibodies	Anti-GQ1b IgG (+)	Anti-GQ1b IgG (-)	
	Serum total IgG	51.9 g/L↑↑	10.70 g/L	
Normal reference range: Gangliosides antibodies (-); serum total IgG (8.60–17.40 g/L). IgG, immunoglobulin G.				

were absent. Electromyography (EMG) indicated significant motor axon involvement with mild demyelination (Table S1). Serum testing revealed anti-GD1a IgG (++) and anti-GD3 IgG (+++), leading to a diagnosis of severe GBS with the AMAN subtype.

Upon admission, the patient received 0.4 mg/kg of IVIg for 5 days; however, his weakness and dyspnea worsened. On the following day, he became drowsy, and limb muscle strength deteriorated to grade 0. Arterial blood gas analysis showed a PaCO₂ of 97.8 mmHg and a pH of 7.187. Noninvasive mechanical ventilation was initiated but achieved only 90% peripheral oxygen saturation. By day 3, the patient entered a shallow coma with oxygen saturation dropping to 82%, necessitating endotracheal intubation and invasive mechanical ventilation. After a 5-day course of IVIg, the patient's level of consciousness improved to a drowsy state, and limb strength increased to grade 3 (Figure 1(a)). Given the severity of his condition, efgartigimod (10 mg/kg) treatment was initiated on day 6 following the

IVIg course. By day 3 of efgartigimod treatment, the patient could stand unaided, and by day 7, he could walk unassisted. After completing three doses of efgartigimod, the patient was discharged, able to walk 200 m independently, although he could not raise his arms above shoulder level. A follow-up 1 month later showed significant improvement in upper limb function, and blood tests indicated positive anti-GD3 IgG (+).

Case 2

A 54-year-old male patient presented to the emergency department with dizziness, unsteady gait, projectile vomiting, cold sweats, and significantly elevated blood pressure. He also reported limb weakness, facial numbness, diplopia, and blurred vision. Notably, he had recently recovered from a cold and received symptomatic treatment 10 days prior to admission. Neurological examination revealed dysarthria, ophthalmoplegia with horizontal nystagmus, mild peripheral facial paralysis, an absent gag reflex, and MRC grade 4 muscle strength in all limbs. Additional findings included

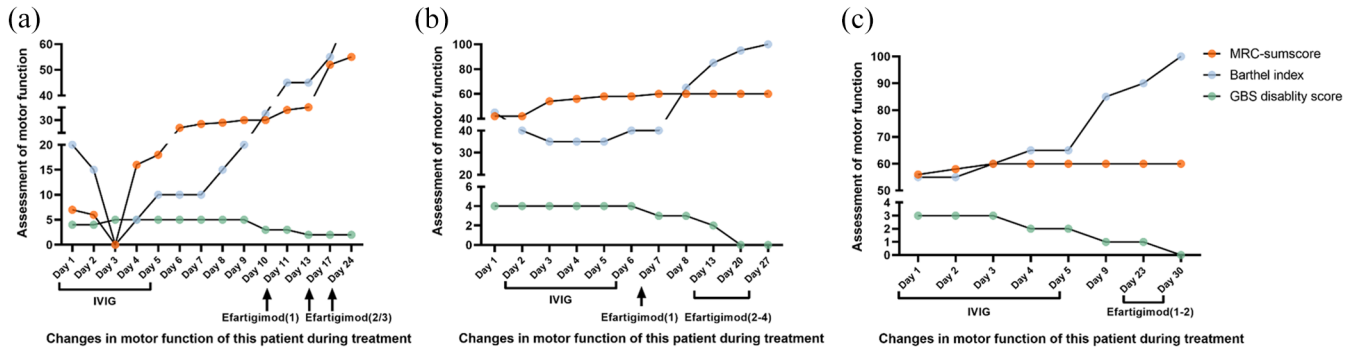


Figure 1. Changes in GBS patients motor function during Efgartigimod treatment. (a) Case 1. (b) Case 2. (c) Case 3. The X-axis indicates the number of days after disease onset. The Y-axis indicates scores from various scales, including the ADL scale, the GBS disability score, and the Barthel index. Lower scores on the GBS disability score indicate less severe disability in GBS patients, whereas higher scores on ADL and the Barthel index indicate better motor function. ADL, activities of daily living; GBS, Guillain-Barré syndrome.

normal muscle tone, absent tendon reflexes, and decreased sensation in a stocking-glove distribution. The patient was unable to adequately perform the finger-nose and heel-shin tests and could not cooperate during the gait assessment. A lumbar puncture performed on admission revealed an opening pressure of 170 mmH₂O, no cells in the CSF, and a slight increase in protein concentration (50 mg/dL). EMG showed predominantly sensory nerve axonal involvement with a decreased F-wave rate (Table S2). CSF analysis tested positive for anti-GT1a IgG (+++) and anti-GQ1b IgG (++), while serum analysis was positive for anti-GT1a IgG (+++), anti-GQ1b IgG (+++), anti-GD3 IgG, and anti-GT1b IgG (+). Based on these findings, the patient was diagnosed with MFS-GBS overlap syndrome due to the presence of limb weakness.¹⁰

On day 2 of admission, the patient was started on IVIg therapy at 0.4 g/kg for 5 days. Despite completing the IVIg course, the patient's dizziness and projectile vomiting persisted. Neurological examinations revealed progressive proximal sensory disturbances in the limbs. As symptoms continued to worsen, a weekly intravenous infusion of efgartigimod (80 mg) was initiated 2 days after the IVIg treatment. After the first dose of efgartigimod, the patient's clinical symptoms stabilized, with no further progression. By day 4 postinitial dose, improvements in blurred vision, ataxia, and sensory disturbances were observed, allowing the patient to stand and walk with assistance. After the second dose, serum anti-ganglioside antibodies remained positive for anti-GT1a IgG (+++) and anti-GQ1b IgG (+++). By the

third dose, most symptoms, including facial paralysis, dizziness, and sensory deficits, resolved, enabling the patient to walk independently, although some residual limitations in eye movement persisted. By the fourth dose, all symptoms had resolved, marking this dose as a consolidation therapy. The motor function changes during treatment were shown in Figure 1(b). However, serum anti-GT1a IgG (++) and anti-GQ1b IgG (++) remained positive for over 3 months following clinical resolution.

Case 3

A 27-year-old male patient was admitted with eye muscle paralysis, ataxia, and lower limb weakness, with no recent history of cold or diarrhea. Muscle strength was assessed at MRC grade 5 in the upper limbs and MRC grade 4 in the lower limbs, with normal tone and poor coordination. Deep tendon reflexes were absent, and Babinski's sign was negative bilaterally. A lumbar puncture performed on the day of admission revealed elevated CSF pressure with normal protein and cell counts. Both CSF and serum tested positive for anti-GQ1b antibodies. EMG showed low F-wave conduction velocity in the left median nerve, varied F-wave morphologies in the right median nerve, bilateral ulnar nerves, and bilateral tibial nerves, and an absence of F-wave conduction in the left peroneal nerve (Table S3). The patient was diagnosed with MFS-GBS overlap syndrome.

After completing a 5-day course of IVIg at 0.4 g/kg, the patient showed significant improvement in ataxia and weakness. However,

ophthalmoplegia persisted. Twenty-five days after symptom onset, the patient's eyes remained fixed in position. He then received weekly intravenous efgartigimod (800mg) for 2 weeks. By day 2 of treatment, the patient reported improvement in horizontal eye movement. One week later, both horizontal and vertical movements had improved by approximately 80%. By day 3 post-second dose, there was complete resolution of symptoms, and the serum ganglioside antibody tests reported negative findings. Figure 1(c) illustrated the changes in motor function that occurred during treatment.

Discussion

Our case series found that sequential treatment with efgartigimod is both effective and safe for patients with AMAN-GBS and MFS-GBS overlap syndrome. Efgartigimod may shorten the disease course in severe GBS cases, especially for those with poor responses to IVIg.

All three patients in this study had detectable ganglioside antibodies in their serum or CSF. Case 1, with severe AMAN-GBS, tested positive for both anti-GD1a IgG and anti-GD3 IgG. Case 2, diagnosed with MFS-GBS overlap syndrome, tested positive for anti-GQ1b IgG and anti-GT1a IgG and presented with perioral and facial numbness. Case 3, also with MFS-GBS overlap syndrome, tested positive for anti-GQ1b IgG. These findings are consistent with previous reports linking ganglioside antibodies to clinical phenotypes of GBS.²⁰ All patients experienced significant symptomatic improvement during treatment, with varying reductions in serum ganglioside IgG antibody titers and total IgG levels. Our results supported previous studies that highlight the pathogenic role of anti-ganglioside IgG antibodies in GBS.^{21,22} Notably, Case 1, presenting with the most severe symptoms, had lower serum anti-ganglioside antibody titers than Case 2, whose symptoms were comparatively less severe despite having higher antibody levels. Case 3, with the mildest symptoms, had higher pretreatment total serum IgG levels than Cases 1 and 2. Furthermore, the serum anti-ganglioside IgG antibodies in Case 2 remained strongly positive for over 3 months after clinical resolution. These findings suggested that while anti-ganglioside antibodies are involved in GBS pathogenesis, their levels do not necessarily correlate with symptom severity.²³ The reduction in IgG levels following efgartigimod infusion

may inhibit the inflammatory cascade and complement activation triggered by pathogenic IgG, potentially partly explaining its rapid onset of action in our patients.^{21,22,24}

GBS is a neurological emergency, yet treatment options are currently limited to PE and IVIg.¹ Additionally, over 20% of patients with GBS experience poor outcomes or death, even with timely treatment.¹⁴ In Case 1, standard IVIg therapy was administered for 5 days, starting within 3 days of symptom onset. Despite this, recovery plateaued within 3–5 days after completing IVIg therapy, likely due to rapid IVIg consumption. Efgartigimod was introduced on day 6 after IVIg treatment, and the patient was able to achieve independent ambulation by 19 days post-onset, reducing the hospital stay and risk of complications. This provides evidence supporting the potential of efgartigimod to prevent further nerve damage and halt disease progression in patients with GBS with the AMAN subtype.²⁵ In patients with MFS-GBS overlap syndrome, despite early IVIg treatment, approximately 44% still experience persistent ophthalmoplegia at 6 months, with over 30% suffering from residual sensory disturbances.²⁶ In this study, Cases 2 and 3 were diagnosed with MFS-GBS overlap syndrome.¹⁰ Case 2 showed symptom progression despite IVIg therapy, but stabilization was observed after the first dose of efgartigimod. Complete resolution, including ophthalmoplegia, was achieved within 1 month after four doses of efgartigimod. Case 3 experienced a 'washout' period of up to 24 days after completing IVIg therapy,²⁷ during which ophthalmoplegia persisted. However, following two doses of efgartigimod, the patient's eye movement returned to normal. These cases suggest that efgartigimod may shorten the duration of AMAN-GBS and MFS-GBS overlap syndrome and may offer a favorable therapeutic effect on ophthalmoplegia, although the possibility of spontaneous recovery or the contribution of IVIg could not be completely excluded.

A previous study indicated that efgartigimod alone also is effective in treating GBS, making it a potential new treatment option.¹⁹ However, given the limited current research on efgartigimod for GBS, further studies are needed to clarify the specific mechanisms underlying this therapy.¹⁹ A phase II randomized, double-blind, controlled clinical trial (NCT05701189) is currently

undergoing, which compares the efficacy and safety of efgartigimod and IVIg as independent treatments for GBS; the study's findings may provide further insights.

In conclusion, although our case series suggests that sequential treatment with efgartigimod may effectively shorten the duration of GBS, particularly in cases with poor responses to IVIg, further research is needed to validate these findings.

Declarations

Ethics approval and consent to participate

All patients were satisfied with their diagnosis and treatment and provided written informed consent. The study was approved by the ethics review committee of West China Hospital of Sichuan University (Approval number: 2023 (1960)).

Consent for publication

All patients provided a written informed consent to the publication of this article.

Author contributions

Sihui Chen: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Writing – original draft.

Ruwei Ou: Methodology; Writing – review & editing.

Qianqian Wei: Methodology; Writing – review & editing.

Bi Zhao: Data curation; Writing – review & editing.

Xueping Chen: Conceptualization; Methodology; Project administration; Resources; Writing – review & editing.

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Competing interests

The authors declare that there is no conflict of interest.

Availability of data and materials

The data that support the findings of this study are available within the article.

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Supplemental material

Supplemental material for this article is available online.

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